Algebraic Geometry – Exercise Sheet 1

Week 1: Affine algebraic sets, projective space and localisation

Monday and Thursday afternoon sessions; Lectures 1–5

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lecture 1 only

Exercise 1. Three points in the affine plane $\star$ CORE

Let
$$S = \{(0,0),(1,0),(0,1)\} \subset \Bbbk^2.$$

(i) Show that
$$\mathbf{I}(S) = (x,y) \cap (x-1,y) \cap (x,y-1).$$

(ii) Prove directly that
$$\mathbf{I}(S) = \big(xy,\, x(x-1),\, y(y-1)\big).$$

(iii) Show that evaluation at the three points induces an isomorphism
$$\Bbbk[x,y]/\mathbf{I}(S) \simeq \Bbbk^3.$$

Exercise 2. Two axes and their coordinate rings $\star$ CORE

In $\Bbbk^2$, set $X = \mathbf{V}(x)$ and $Y = \mathbf{V}(y)$.

(i) Show that
$$X \cup Y = \mathbf{V}(xy), \qquad X \cap Y = \mathbf{V}(x,y).$$

(ii) Compute the coordinate rings of X, Y, $X \cup Y$, and $X \cap Y$.

(iii) Explain geometrically why $\Bbbk[x,y]/(xy)$ has zero divisors.

Exercise 3. The parabola and the cusp $\star\star$ CORE

Consider
$$P = \mathbf{V}(y - x^2), \qquad C = \mathbf{V}(y^2 - x^3) \subset \Bbbk^2.$$

(i) Show that $t \mapsto (t,t^2)$ defines an isomorphism $\Bbbk \to P$.

(ii) Show that $t \mapsto (t^2,t^3)$ defines a morphism $\nu : \Bbbk \to C$ which is bijective on the underlying sets.

(iii) Write the corresponding homomorphism of coordinate rings and prove that ν is not an isomorphism.

Exercise 4. Automorphisms of the affine line $\star\star$ CHOICE

Assume in this exercise that $\Bbbk$ is infinite.

(i) Let $F : \Bbbk \to \Bbbk$ be a morphism of affine algebraic sets. Show that there is a unique polynomial $f \in \Bbbk[t]$ such that $F(a) = f(a)$ for every $a \in \Bbbk$. If u denotes the coordinate on the target, describe the pullback
$$F^* : \Bbbk[u] \longrightarrow \Bbbk[t].$$
If $G : \Bbbk \to \Bbbk$ is a second morphism, use t, u, v as the coordinates on the source, intermediate and target affine lines, respectively, and check explicitly that $(G \circ F)^* = F^* \circ G^*$.

(ii) Prove that F is an automorphism if and only if
$$f(t) = at + b \qquad \text{with } a \in \Bbbk^\times,\ b \in \Bbbk.$$

(iii) Define
$$G = \Bbbk \rtimes \Bbbk^\times \quad \text{by} \quad (b,a)(b',a') = (b + ab', aa').$$
Show that $(b,a) \mapsto (t \mapsto at + b)$ is an isomorphism from G to the group of automorphisms of the affine line, where the group law is composition of maps. Explain why pullback gives an anti-isomorphism with $\mathrm{Aut}_{\Bbbk\text{-alg}}(\Bbbk[t])$, and deduce an explicit isomorphism
$$\mathrm{Aut}_{\Bbbk\text{-alg}}(\Bbbk[t]) \simeq \Bbbk \rtimes \Bbbk^\times.$$

Exercise 5. Equations, zero sets and ideals of subsets ⋆⋆ CORE

Work over $\mathbb{R}$.

(i) Show that, for every $m \geq 2$, the three distinct ideals

$$(x, y), \qquad (x^2, xy, y^3), \qquad (x, y)^m$$

define the same subset $\{(0,0)\} \subset \mathbb{R}^2$. Determine the ideal of polynomials vanishing on this subset.

(ii) Let $C = \{\pm 1\}^n \subset \mathbb{R}^n$ be the set of vertices of the n-dimensional cube. Prove that

$$\mathbf{I}(C) = (x_1^2 - 1, \ldots, x_n^2 - 1).$$

(iii) Regard $\mathbb{Z}$ as a subset of $\mathbb{R}$. Compute $\mathbf{I}(\mathbb{Z}) \subset \mathbb{R}[t]$. Deduce that $\mathbb{Z}$ is not an algebraic subset of the affine line over $\mathbb{R}$.

Exercise 6. Polynomial systems versus linear systems ⋆ CORE

Fix $n, d \geq 1$.

(i) Recall why an invertible linear system of n equations in n unknowns has exactly one solution.

(ii) Over $\mathbb{C}$, determine all solutions of

$$x_1^d = 1, \quad x_2^d = 1, \quad \ldots, \quad x_n^d = 1.$$

Show that there are exactly d^n solutions.

(iii) Write a system of n polynomial equations in n variables having no solution over $\mathbb{R}$.

(iv) Explain precisely which expectations imported from linear algebra have failed in (ii) and (iii).

Thursday afternoon — material available: Lectures 1–4

Exercise 7. Radicals and reduced rings ⋆ CORE

Compute

$$\sqrt{(x^2, y^3)}, \qquad \sqrt{(x^2 y^3)}$$

in $\Bbbk[x, y]$. Decide which of the rings

$$\Bbbk[x]/(x^2), \qquad \Bbbk[x, y]/(xy), \qquad \Bbbk[x, y]/(x^2, xy)$$

are reduced.

Exercise 8. Points at infinity ⋆ CORE

In the affine chart $x_0 \neq 0$ of $\mathbf{P}^2_{\Bbbk}$, write $x = x_1/x_0$ and $y = x_2/x_0$.

(i) Homogenise the parabola $y = x^2$ and determine its points at infinity.

(ii) Homogenise the hyperbola $xy = 1$ and determine its points at infinity.

(iii) Explain geometrically why the answers are different.

Exercise 9. The line through two points of projective space ⋆ CHOICE

Let $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ be distinct points of $\mathbf{P}^n_{\Bbbk}$. Show that the line through them is defined by the 3×3 minors of

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}.$$

Write the equation explicitly for $n = 2$.

Exercise 10. Points in $\mathbf{P}^2$ from minors ⋆⋆ CHOICE

(i) Compute the 2×2 minors of

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}.$$

Determine their common zero set in $\mathbf{P}^2$ and prove that the ideal which they generate is the radical ideal of the three coordinate points.

(ii) Let $\ell_1, \dots, \ell_4$ define four distinct lines in $\mathbf{P}^2$, no three of which are concurrent. Compute the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}.$$

Show first that their common zero set consists of the six pairwise intersection points, and then check locally that the ideal defines each of these points with reduced structure.

Exercise 11. Equations of the Segre and Veronese images ⋆⋆⋆ OPTIONAL

(i) For

$$([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0t_0 : s_0t_1 : s_1t_0 : s_1t_1],$$

compute the kernel of the homomorphism $z_{ij} \mapsto s_it_j$ and deduce an equation for the image in $\mathbf{P}^3$.

(ii) For the quadratic Veronese map $\mathbf{P}^2 \to \mathbf{P}^5$, arrange the six target coordinates as a symmetric 3×3 matrix. Prove that the kernel of $z_{ij} \mapsto x_ix_j$ is generated by the 2×2 minors of this matrix.

Exercise 12. Localising the coordinate axes ⋆ CORE

Let

$$A = \Bbbk[x, y]/(xy).$$

(i) Compute A_x and A_y.

(ii) Compute $A_{(x)}$ and $A_{(y)}$.

(iii) Explain what these localisations retain and what they discard geometrically.

Exercise 13. Localisation with a nilpotent ⋆ CHOICE

Let $A = \Bbbk[\varepsilon]/(\varepsilon^2)$. Compute

$$A_\varepsilon, \qquad A_{(\varepsilon)}, \qquad A_{1+\varepsilon}.$$

Explain why the three answers are so different.

Exercise 14. Two finite module structures ⋆⋆ CHOICE

Let

$$B = \Bbbk[x, y]/(y^2 - x).$$

(i) Prove that the homomorphism $\Bbbk[y] \to B$ is an isomorphism.

(ii) Prove that B is free with basis $(1, y)$ over $\Bbbk[x]$, and free with basis (1) over $\Bbbk[y]$.

(iii) Write the two corresponding morphisms to the affine line on $\Bbbk$-points. After extending the field to an algebraic closure, compare their fibres; distinguish the cases $\mathrm{char}(\Bbbk) \neq 2$ and $\mathrm{char}(\Bbbk) = 2$.

Exercise 15. Noether normalisation for the hyperbola ⋆⋆⋆ CHOICE

Let

$$B = \Bbbk[x, y]/(xy - 1), \qquad u = x + y \in B.$$

(i) Show that u is transcendental over $\Bbbk$, so that $\Bbbk[u] \subset B$ is a polynomial subring.

(ii) Prove that B is generated by 1 and x as a $\Bbbk[u]$-module. In particular, B is finite over $\Bbbk[u]$.

(iii) Let $H = \mathbf{V}(xy - 1) \subset \Bbbk^2$ and consider

$$\pi : H \longrightarrow \Bbbk, \qquad (a, b) \longmapsto a + b.$$

For $\lambda \in \Bbbk$, identify $\pi^{-1}(\lambda)$ with the roots in $\Bbbk$ of a quadratic polynomial. Deduce that every fibre has at most two points, and that it is non-empty when $\Bbbk$ is algebraically closed.

Exercise 16. Automorphisms of the rational function field ⋆⋆⋆ OPTIONAL

Let $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ belong to $\mathrm{GL}_2(\Bbbk)$ and define

$$\tau_M : \Bbbk(t) \longrightarrow \Bbbk(t), \qquad t \longmapsto \frac{at + b}{ct + d}.$$

(i) Prove that τ_M is a $\Bbbk$-automorphism, that scalar multiples of M give the same automorphism, and that

$$\tau_M \circ \tau_N = \tau_{NM}.$$

(ii) Using the standard fact $[\Bbbk(t) : \Bbbk(r)] = \deg(r)$ for every non-constant rational function $r \in \Bbbk(t)$, prove that

$$\mathrm{PGL}_2(\Bbbk) \longrightarrow \mathrm{Aut}_{\Bbbk\text{-alg}}(\Bbbk(t)), \qquad [M] \longmapsto \tau_{M^{-1}},$$

is an isomorphism.

(iii) Determine the subgroup of automorphisms which preserves the subring $\Bbbk[t] \subset \Bbbk(t)$, and relate it to Exercise 4.

After Friday's lecture or independent practice

Exercise 17. A presheaf which is not a sheaf $\star$ **OPTIONAL**

Let $X = U \amalg V$ be the disjoint union of two non-empty open subsets. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(W) = \mathbb{Z} \quad \text{for } W \neq \varnothing, \qquad \mathcal{F}(\varnothing) = 0,$$

with the evident restriction maps. Show that $\mathcal{F}$ is not a sheaf.

Exercise 18. Continuous and holomorphic functions $\star\star$ **CHOICE**

Let $U \subset \mathbb{C}$ be connected and contain at least two points. Show that $\mathcal{O}(U)$ has no zero divisors, whereas $\mathcal{C}^0(U)$ has non-zero functions f, g with $fg = 0$.

Exercise 19. $\mathcal{O}(d)$ on the projective line $\star\star\star$ **OPTIONAL**

Identify $\mathbf{P}^1_{\mathbb{C}}$ with $\mathbb{C} \cup \{\infty\}$. Define $\mathcal{O}(d)$ as the sheaf of meromorphic functions which are holomorphic away from ∞ and have at most a pole of order d there when $d \geq 0$; for $d < 0$, require a zero of order at least $-d$ at ∞. Compute its global sections.